The printed almost-simple chiral atlas: complete applications and its domain

Private L070 recovery component

The fixed source is Hartley–Hubard–Leemans, *Two atlases of abstract chiral polytopes for small groups*, Ars Math. Contemp. 5 (2012), 371–382. We distinguish its 79 printed group models, their 88 occurrences in Tables 2–7, and the larger domain stated in Section 4. All computations below use those exact named models; printed polytope counts are never search filters. The separate first-atlas classification is not repeated here.

Convention and retained recognition. An admissible prefix marking is an ordered tuple

$$\alpha = (1, \alpha_1, \ldots, \alpha_{r-1})$$

generating G, with the adjacent orders greater than two, nonadjacent differences involutions, and the full intersection property for the subgroups generated by $\alpha_i^{-1}\alpha_j$ on each index set. The acquired recognition theorem gives the corresponding regular or chiral abstract polytope. A marking is regular exactly when the simultaneous inversion of $\alpha_1, \ldots, \alpha_{r-1}$ extends to an automorphism. Markings are identified by ordered generator images under $\mathrm{Aut}(G)$; reversal is not additionally identified. In the chiral part the source counts the two enantiomorphic forms separately: Sections 3 and 4 give the ordered-marking construction, and Section 6 expressly divides its count by two only when those forms are to be identified. This convention was independently checked in the acquired source-orientation review.

Theorem 1. *Relative to the acquired named models, recognition and exhaustive-prefix theorems, and the stated finite group-operation implementations, all chiral columns for the 79 printed groups are determined. Their total is*

$$16974 = 15056 \text{ (rank 3)} + 1890 \text{ (rank 4)} + 28 \text{ (rank 5)}.$$

Of the 88 printed rank distributions, 87 agree. Table 6, page 381, prints 26 for $\mathrm{PSU}_4(2)$*; the correct rank-three count in the same convention is 46. There are no other rank-count discrepancies in the printed chiral rows.*

Proof. The accepted prefix theorem enumerates inner-conjugacy classes by choosing one representative of each nonidentity, noninvolutory first conjugacy class, and then representatives under the centralizer of the prefix subgroup at every extension. Its full intersection test and reflection test are unchanged. Every extension strictly increases that subgroup, so the finite enumeration exhausts all ranks, rather than stopping at the largest rank printed in the source.

The actual executions consist of 72 clean complete whole-group calls, one complete 25-class union for S_9, and six complete fixed-plan unions for $L_3(4).D_{12}$, $P\Sigma U_3(5)$, J_2, $L_3(5) : 2$, $P\Gamma U_3(5)$ and $M_{22} : 2$. The last six replace six earlier nonreturns, which receive no census credit. Each fixed-plan call checks its full pinned first-representative list against freshly computed complete classes by actual conjugating transporters. The receipt union requires the exact same literal list, model, engine and acceptance, and each index exactly once. In particular an index from a different randomly ordered class list cannot be substituted.

The application reader reconstructs the whole-call drivers and exact input/output inventories, checks all ordered class-return markers and the terminal census marker, and requires a clean terminal process. For class unions it invokes the unchanged accepted v2 or v3 receipt verifier in isolated Python processes. The fresh application makes no group calls. All literal census files equal their previously acquired unions. The older whole-call receipts bind their exact program, model and driver, but record only the GAP executable locator, not a complete runtime-file trace; that historical provenance limit is retained explicitly. The stronger v2/v3 runtime inventories are not retroactively attributed to them.

Finally $\mathrm{Out}(G)$ acts freely on inner classes of generating markings. Indeed, an automorphism stabilizing an inner class agrees on every generator with an inner automorphism, hence equals it. Regularity, rank and ordered type are invariant. Thus each returned rank/type/reflection stratum has size divisible by $|\mathrm{Out}(G)|$,

and division gives exactly the ordered $\mathrm{Aut}(G)$ classes. The reader checks these divisions before joining all 79 groups to all 88 source occurrences. The resulting complete machine-readable row list gives the stated total and sole discrepancy. The previously accepted independent literal 46-class certificate for $\mathrm{PSU}_4(2)$ corroborates this row without assuming exhaustion or outer-index division; it is acquired separately, not counted twice. □

The Section 6 aggregates. The accepted regular-column application gives 9210 regular classes, with rank distribution $8319, 810, 61, 16, 3, 1$ for ranks 3 through 8. Its only row correction is $\mathrm{PSL}_2(27)$, 27 to 26. Combining that result with the theorem gives the following family totals; overlapping group occurrences remain in their individual families.

Source table	Regular	Chiral	Regular rank ≥ 4	Chiral rank ≥ 4
2	1536	4780	271	360
3	804	1710	234	348
4	3903	1882	45	338
5	1846	3998	180	448
6	1202	4242	170	464
7	14	412	0	0

The nine duplicate occurrences contribute 95 regular and 50 chiral classes and are removed only in the global sum. The source's table-total arithmetic errors, already visible without a census, are Table 3 chiral $1702 \to 1710$, Table 4 chiral $2050 \to 1882$, and Table 5 regular $1840 \to 1846$. The two newly verified row corrections then give the global $9205 \to 9210$ regular and $17114 \to 16974$ chiral totals. The Section 6 comparison for the PSL_2 family has rank-at-least-four counts 45 regular and 338 chiral, rather than 51 and 506.

Mirror reversal is a fixed-point-free involution on the chiral ordered classes: a fixed class would provide the reflection automorphism excluded by chirality. Therefore identifying enantiomorphic forms gives $16974/2 = 8487$ classes, by rank $7528, 945, 14$. This does not identify duals. Under this convention regular classes still outnumber mirror pairs, whereas the two oriented forms counted separately outnumber regular classes. The qualitative sporadic, unitary and Suzuki comparisons in Section 6 remain true. Its older 5265 count is explicitly up to both isomorphism and duality, and the Conder genus counts are attributed historical context from other work; neither is inferred from this census. The source excludes $\mathrm{Aut}(J_2)$ from its regular aggregate, as do our 79 printed models. $J_2 : 2$ has order 1209600 and lies outside both numerical cutoffs discussed below.

Lemma 1 (Automorphisms of an almost simple preimage)**.** *Let S be nonabelian finite simple and $S \leq G \leq \mathrm{Aut}(S)$. Then restriction identifies*

$$\mathrm{Aut}(G) \cong N_{\mathrm{Aut}(S)}(G), \qquad N_{\mathrm{Aut}(S)}(G)/S = N_{\mathrm{Out}(S)}(G/S).$$

Proof. The centralizer of S in $\mathrm{Aut}(S)$ is trivial. Any nontrivial normal subgroup of G contains S: otherwise its commutator with S is trivial, which places it in that centralizer. Hence S is the unique minimal normal subgroup and is characteristic. An automorphism of G restricts to $\beta \in \mathrm{Aut}(S)$. Its value at $g \in G$, considered as an automorphism of S, must be $\beta g \beta^{-1}$, by conjugation on S. It therefore embeds in the displayed normalizer. Conversely a normalizer element induces an automorphism of G by conjugation. The same trivial centralizer makes this action faithful. The last equality follows since G is the full preimage of G/S. □

The group-order and automorphism-name columns. The accompanying 79-model registry rejoins every actual ordered generator array to its accepted named-model selector; natural alternating and symmetric models are identified by permutation parity and their checked full factorial order. These are identifications, not naming by order alone. Two printed group orders are wrong: M_{22} is 443520, not 443510, and $\mathrm{PSL}_2(25).2$ is 15600, not 7800.

For automorphism names we retain the full named simple-group automorphism classifications and the accepted outer-preimage identifications. The classified-input locators and the exact named target model for every row are recorded separately. Except for $L_3(4)$ and $U_3(5)$ the relevant outer groups are abelian, so every outer subgroup has full normalizer. In $D_{12} \cong \mathrm{Out}(L_3(4))$, the central order-two subgroup is normal, each reflection subgroup has a four-group normalizer, and that four-group is self-normalizing. The other relevant subgroups are normal. The already accepted ATLAS-word joins distinguish the central 2_1, field 2_2 and graph 2_3 types. Thus precisely the two reflection preimages and their four-group preimage

have automorphism group $L_3(4).2^2$; the others have the full $L_3(4).D_{12}$. In $S_3 \cong \mathrm{Out}(U_3(5))$ the reflection preimage $P\Sigma U_3(5)$ is self-normalizing; the other three preimages have full automorphism group $P\Gamma U_3(5)$. All these small normalizers are computed literally in the accompanying reader.

The source's $\mathrm{Aut}(\mathrm{PSU}_3(3)) = P\Gamma L_3(3)$ is a label error: the retained unitary automorphism theorem gives $P\Gamma U_3(3)$, of order 12096. In contrast, $P\Gamma L_3(3) = PGL_3(3)$ has order

$$\frac{(3^3-1)(3^3-3)(3^3-3^2)}{3-1} = \frac{11232}{2} = 5616,$$

since $\mathbb{F}_3$ has no nontrivial field automorphism. For J_2, the acquired full action is the accepted pair PrimitiveGroup(100, 1) inside PrimitiveGroup(100, 2). A saved six-letter word in the latter generators gives an actual outside involution; a clean bounded verification checks the exact arrays, normalization, index two and generation. Thus the printed colon $J_2 : 2$ has an actual complement, not merely an outer quotient of order two. The unsuccessful initial length-two proposal and the longer proposal's GAP parse warning are preserved; only the clean fixed-witness verification supplies this join.

Theorem 2 (Historical-domain boundary, with classified inputs)**.** *Using the retained CTblLib transcription of the 1985 ATLAS contents and the accepted complete simple/outer roster below one million, the Section 4 socle condition gives 80 almost simple isomorphism classes of order below one million. Exactly 79 have order below 900000, and their named models are precisely the 79 printed groups. The additional class is* $\mathrm{Sp}_4(4)$ *of order 979200.*

Proof. The accepted L054 domain theorem gives every almost simple group below one million as a preimage of an outer subgroup. Its unique-minimal-normal argument proves that isomorphisms between such preimages are exactly outer conjugacies. Its finite outer-subgroup lists and complete simple roster are acquired, with their explicit classification premises.

The pinned CTblLib `AtlasPages` record has 97 entries. Its own documentation identifies the original book's page-v contents and later additions. Exactly 93 entries have a nonempty ordinary-ATLAS page list; the other four are later additions $L_2(49), L_2(81), L_6(2), S_{10}(2)$. Retaining only the original ordinary-page entries intersects the accepted small-order roster in 32 socles and 80 almost simple classes. The literal table identifiers for the 79 acquired models are distinct and are exactly this list minus `sp44`. Their checked orders are all below 900000. The remaining group has order 979200 and ordinary ATLAS page 44. This proves both strict-cutoff assertions without guessing an old website edition or using polytope counts to identify groups. □

The precise source limit. The abstract on page 371 says that Γ has order less than 900000 and is listed in the ATLAS. Section 4, page 375, instead explicitly uses a socle S appearing there and $|\Gamma| < 10^6$. Our theorem implements that explicit socle condition. The 79 printed models agree with its 900000 restriction, but not with its one-million restriction. We have not inspected a scan of the 1985 contents: the pinned CTblLib transcription is an openly retained premise. We do not assert that every extension has its own separately printed 1985 table. The order boundary and this Γ-versus-S wording distinction are kept separate. No polytope census for $\mathrm{Sp}_4(4)$ is supplied by this component; no conclusion about an omitted zero count is licensed. Consequently this application closes the printed second-atlas entries and their aggregate corrections, but does not silently convert the larger Section 4 assertion into the narrower printed roster.

Attribution and limits. Root authored the prefix algorithm/theory, most original whole-group applications and the fixed queue, and suggested the normalizer lemma. The present author wrote the v2/v3 receipt repair, ran the four assigned fixed-plan queues, and assembled the present exact joins and small normalizer calculations. The corresponding independent reviews are acquired with these roles unchanged. The regular application and U4(2) positive correction have their separate independent reviews. Publication provided a bounded Section 6 source-only reading. No new polytope census, whole-manuscript acceptance, open-question resolution, Lean proof or compression measurement is asserted here.